9 Databases

In this chapter, you will learn about:
- ★ the design and use of single table databases
  - fields
  - records
  - validation
- ★ the use of basic data types
  - text/alphanumeric
  - character
  - Boolean
  - integer
  - real
  - date/time
- ★ the identification, purpose and use of primary keys
- ★ SQL scripts, used to query data stored in a single table database.

9.1 Databases

9.1.1 Single-table databases

A **database** is a structured collection of data that allows people to extract information in a way that meets their needs. The data can include text, numbers, pictures; anything that can be stored in a computer. Relational databases will be studied at A Level but for IGCSE only single-table databases will be studied.
A **single-table database** contains only one table.

Why are databases useful?

Databases prevent problems occurring because:

- if any changes or additions are made it only has to be done once – data is consistent
- the same data is used by everyone
- data is only stored once in relational databases which means no data duplication.

What are databases used for?

To store information about people, for instance:

- patients in a hospital
- pupils at a school.

To store information about things, for instance:

- cars to be sold
- books in a library.

To store information about events, for instance:

- hotel bookings
- results of races.

Find out more

Find five more uses for databases, and for each one decide what sort of information is being stored.

Fields and records – the building blocks for any database

Inside a database, data is stored in **tables**, which consists of many **records**. Each record consists of several **fields**. The number of records in a table will vary as new records can be added and deleted from a table as required. The number of fields in a table is fixed so each record contains the same number of fields.

An easy way to remember this is: each record is a row in the table and each field is a column in the table.

Note: while databases can contain multiple tables, all the databases considered in this chapter will contain a single table.

Table

Record 1	Field 1	Field 2	Field 3	Field 4
Record 2	Field 1	Field 2	Field 3	Field 4
Record 3	Field 1	Field 2	Field 3	Field 4
Record 4	Field 1	Field 2	Field 3	Field 4
Record 5	Field 1	Field 2	Field 3	Field 4
Record 6	Field 1	Field 2	Field 3	Field 4

Row

Column

▲ **Figure 9.1** Structure of a database table

A table contains data about one type of item or person or event, and will be given a meaningful name, for example:

- a table of patients called **PATIENT**
- a table of books called **BOOK**
- a table of doctor's appointments called **APPOINTMENT**.

Each record within a table contains data about a single item, person or event, for example:

- Winnie Sing (a hospital patient)
- IGCSE Computer Science (a book)
- 15:45 on January 2020 (an appointment).

As every record contains the same number of fields, each field in a record contains a specific piece of information about the single item, person or event stored in that record. Each field will have a meaningful name to identify the data stored in it, for example:

- For a hospital patient the fields could include:
  - The patient's first name field called **FirstName**
  - The patient's family name field called **FamilyName**
  - The patient's date of admission field called **DateOfAdmission**

- The name of the patient's consultant field called **`Consultant`**
- The patient's ward number field called **`WardNumber`**
- The patient's bed number field called **`BedNumber`**, etc.

The PATIENT table structure could look like this:

PATIENT Table

Record 1	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 2	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 3	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 4	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 5	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 6	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber

▲ **Figure 9.2** Structure of the PATIENT table

- For the table called BOOK the fields could include:
  - Title of the book called **`Title`**
  - Author of the book called **`Author`**
  - **`ISBN`**, etc.

Activity 9.1

State what fields would you expect to find in each record for the doctor's appointments and give each field a suitable name.

Note: Field names should be a single word, which should not contain any spaces, for example: **`BedNumber`**

Validation

Link

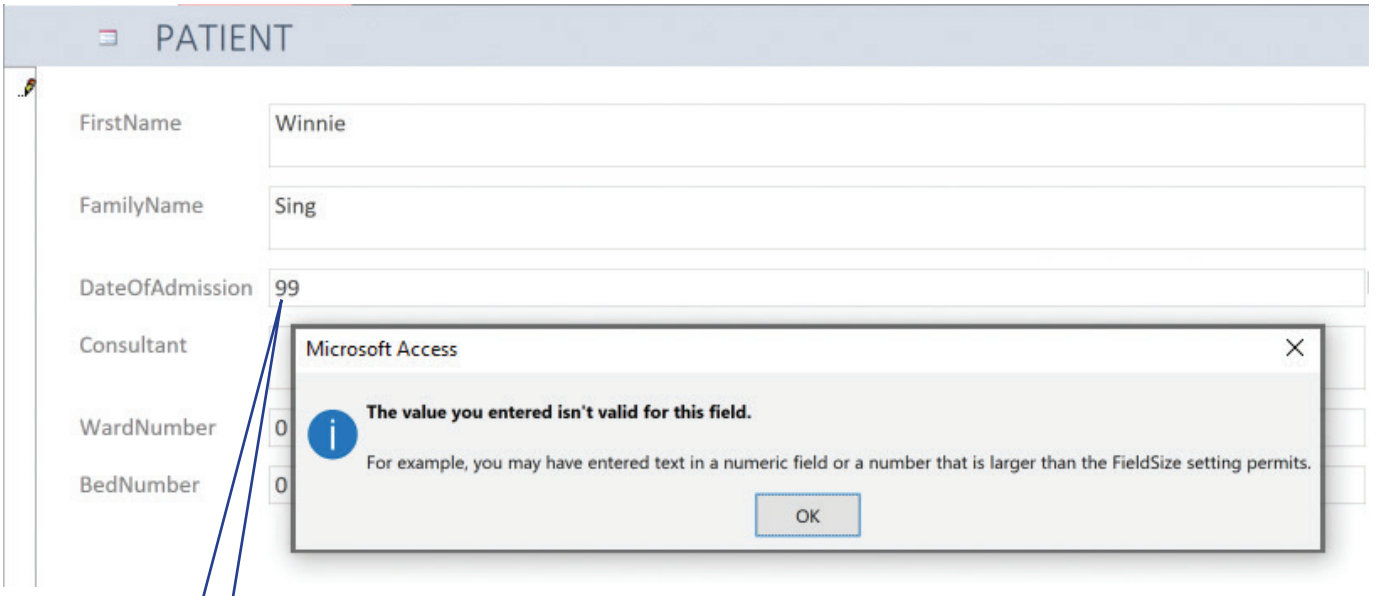
For more on validation see Section 7.5.

The role of validation was discussed in Section 7.5. It may be worth the reader revisiting this part of the book before continuing with this chapter.

Some validation checks will be automatically provided by the database management software that is used to construct and maintain the database. Other validation checks need to be set up by the database developer during the construction of the database.

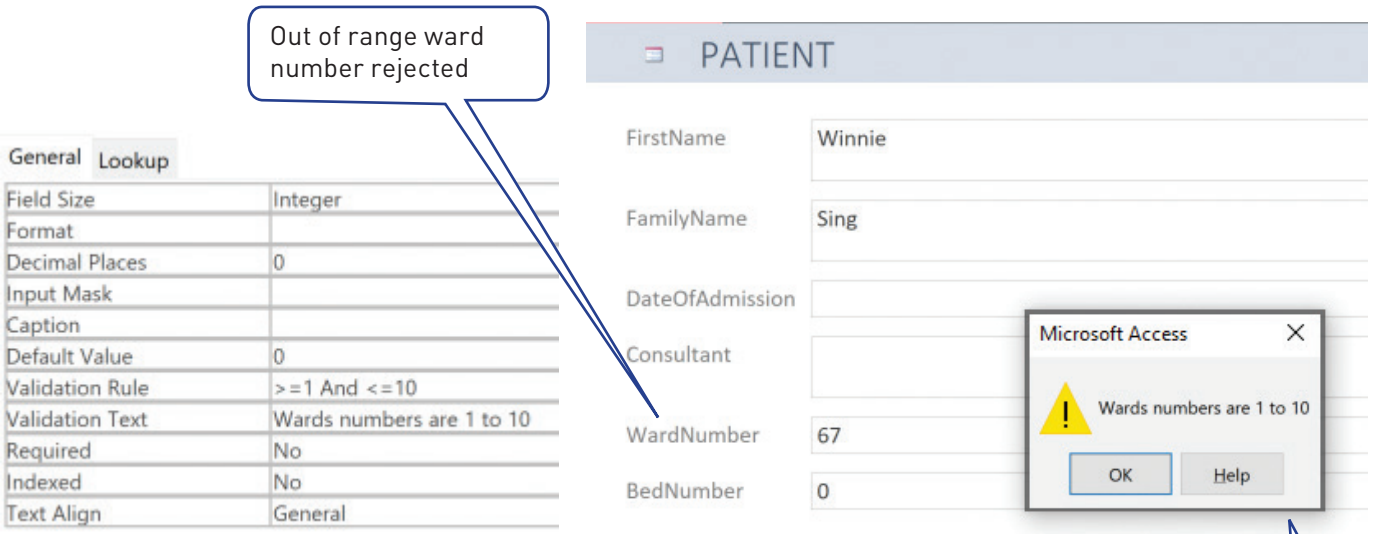
The practical use of a database management system is strongly recommended for all students. Practical examples will be used throughout this chapter. The database management software used is *Microsoft Access 365* as Microsoft Access is used by most schools with students studying IGCSE Computer Science.

For example, the **`DateOfAdmission`** field will automatically be checked by the software to make sure that any data input is a valid date before it can be stored in the PATIENT table.



▲ **Figure 9.3** Automatic validation for entering `DateOfAdmission` in the PATIENT table

However, the `WardNumber` field validation needs to be set up to allow only values 1 to 10 to be entered. This task needs to be completed by the database developer before the database is used.



▲ **Figure 9.4** Validation rule for entering `WardNumber` in the PATIENT table

9.1.2 Basic data types

There are six basic data types that you need to be able to use in a database:

- text/alphanumeric
- character
- Boolean
- integer
- real
- date/time.

What is a data type?

Each field will require a **data type** to be selected. A data type classifies how the data is stored, displayed and the operations that can be performed on the stored value. For example, a field with an integer data type is stored and displayed as a whole number and the value stored can be used in calculations.

These database data types are specified in the syllabus. They are available to use as *Access* data types, but the names *Access* uses may be different from the terms in the syllabus.

Syllabus data type	Description	Access data type
text/alphanumeric	A number of characters	short text/long text
character	A single character	short text with a field size of one
Boolean	One of two values: either True or False, 1 or 0, Yes or No	Yes/No
integer	Whole number	number formatted as fixed with zero decimal places
real	A decimal number	number formatted as decimal
date/time	Date and/or time	Date/Time

Find out more

Using Access (or any other suitable database management software) find all the different date and time formats that are available to use.

Activity 9.2

Using Access (or any other suitable database management software), set up a single table database to store the PATIENT records using the fields given previously:

`FirstName`, **`FamilyName`**, **`DateOfAdmission`**, **`Consultant`**, **`WardNumber`**, **`BedNumber`**.

- Choose a suitable data type for each field.
- Include validation checks to make sure that the ward number has whole number values from 1 to 10, and that the bed number has whole number values from 1 to 8.
- Enter the following data in the PATIENT table:

Winnie Sing, 12/10/2022, Mr Smith, 6, 8

Steve Chow, 23/10/2022, Miss Abebe, 6, 3

Chin Wee, 30/10/2022, Mr Jones, 7, 1

Min Hoo, 1/11/2022, Mr Smith, 6, 1

Peter Patel, 12/11/2022, Mr Jones, 7, 8

Sue Sands, 19/11/2022, Miss Abebe, 6, 2

Farouk Khan, 22/11/2022, Mr Jones, 7, 4

Ahmad Teo 22/11/2022, Mr Jones, 7, 2

You may want to look at the instructions on how to set up the CubScout database later in this chapter if you have not set up a database before.

9.1.3 Primary keys

As each record within a table contains data about a single item, person, or event, it is important to be able to uniquely identify this item. In order to reliably identify an item from the data stored about it in a record there needs to be a field that uniquely identifies the item. This field is called the **primary key**.

A field that is a primary key must contain data values that are never repeated in the table.

The primary key can be a field that is already used, provided it is unique, for example the ISBN in the book table. The PATIENT table would need an extra field for each record as all of the existing fields could contain repeated data. To create a primary key, we could add a new field to each record, for example a unique number could be added to each patient's record. The extra field is:

- Primary key field called **`HospitalNumber`**

Activity 9.3

Using *Access* (or any other suitable database management software), add the **`HospitalNumber`** field to the single table database PATIENT.

- Choose 'text' as the data type for this field.
- Include validation checks to ensure that 8 characters must be entered starting with **`HN`** followed by **`6 digits`** for example **`HN123456`**.
- Choose suitable data to store in each primary key field for the 6 patients in the table.

Activity 9.4

1. Write down **four** database data types. Describe each data type and give an example of a field where it would be a suitable choice.
2. Choose suitable words/phrases from the following list to correctly complete the paragraph that follows:

Word list:	database	fields	records
	record	table	primary key
	data type	validation	text
	char	Boolean	field

You will need to use some words more than once.

A single .... contains one ..... .

Each ..... consists of many ..... . Every .....

has the same number of ..... . Every ..... is given a

..... . Examples of data types are ....., ...

and ..... . Some ..... will have ..... rules.

9.1.4 SQL

Structured Query Language (SQL) is the standard query language for writing scripts to obtain useful information from a database. We will be using SQL to obtain information from single-table databases. This will provide a basic understanding of how to obtain and display only the information required from a database. SQL is pronounced as es-queue-el.

Activity 9.5

1 Using the single table database PATIENT you have created.
   a Write an SQL query to list all Mr Jones' patients.
   b Write an SQL query to list all the patients not in ward 6.
   c Write an SQL query to list all the patients who arrived on 12/11/2022.
   d Write an SQL query to list all the patients who arrived between 12/10/2022 **AND** 30/10/2022.
2 Write down the output from this SQL query.

```
SELECT FirstName, FamilyName, BedNumber
FROM PATIENT
WHERE WardNumber = 7;
```

Activity 9.6

1 Using the single table database PATIENT you have created.
   a Write an SQL query to count the number of patients in ward 7.
   b Write an SQL query to count the number of patients not in ward 7.
2 Write down the output from this SQL query.

```
SELECT HospitalNumber, FirstName, FamilyName, Consultant
FROM PATIENT
ORDER BY Consultant, FamilyName;
```

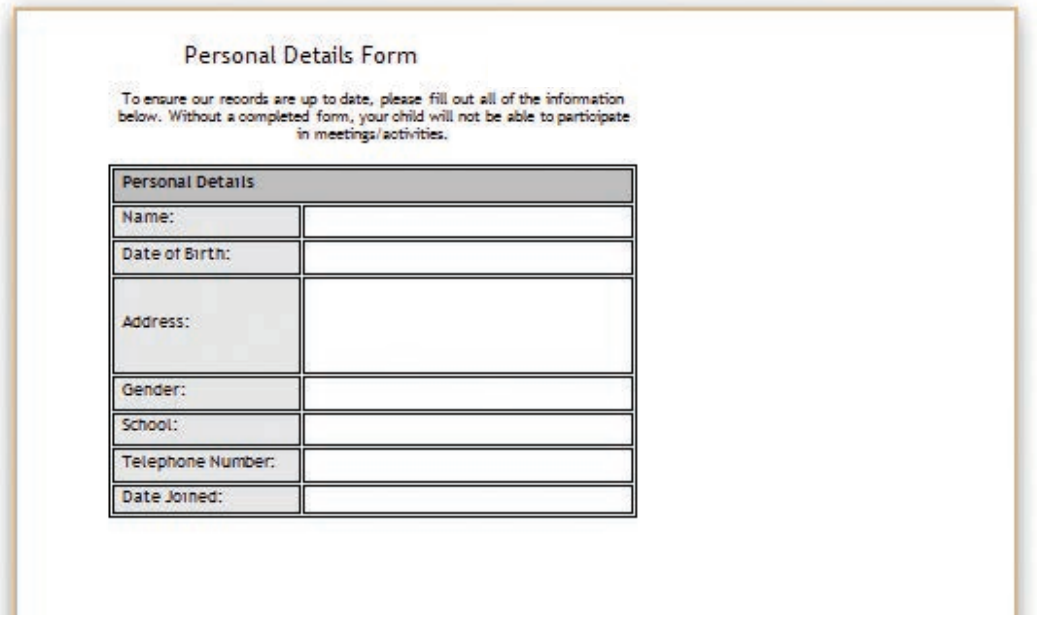
Practical use of a database

As an IGCSE Computer Science student you need to be able to do the following:

- define a single-table database from given data storage requirements
- choose a suitable primary key for a database table
- read, complete and understand SQL scripts.

In order to do this, you will need to use a database management system. The following case study shows how to set up a database with *Microsoft Access 365* and complete the tasks described above.

Boys and girls between the ages of seven and eleven can join a cub scout group. (http://en.wikipedia.org/wiki/Cub_Scout). Each cub scout group needs to keep records about its members. Most groups will keep the following information about each cub in their group:



Personal Details Form

To ensure our records are up to date, please fill out all of the information below. Without a completed form, your child will not be able to participate in meetings/activities.

Personal Details	
Name:	
Date of Birth:	
Address:	
Gender:	
School:	
Telephone Number:	
Date Joined:	

▲ **Figure 9.7** Enrolment form

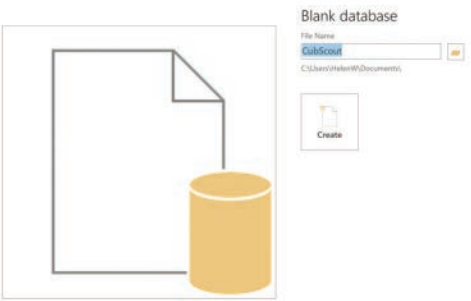
Define a single-table database from given data storage requirements and choose a suitable primary key

To create the cub scout database, open *Access* and select the `Blank database` template.



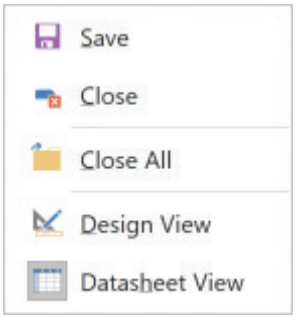
▲ **Figure 9.8** Blank database template

Then type the Filename `CubScout` and click the create button.



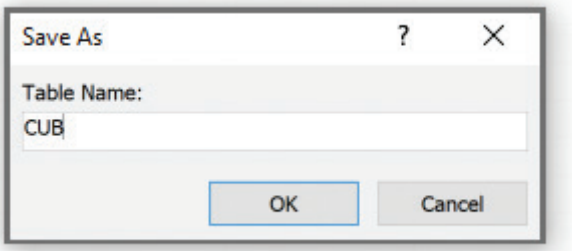
▲ **Figure 9.9** Creating the CubScout database

Select the table design view...



▲ **Figure 9.10** Design view

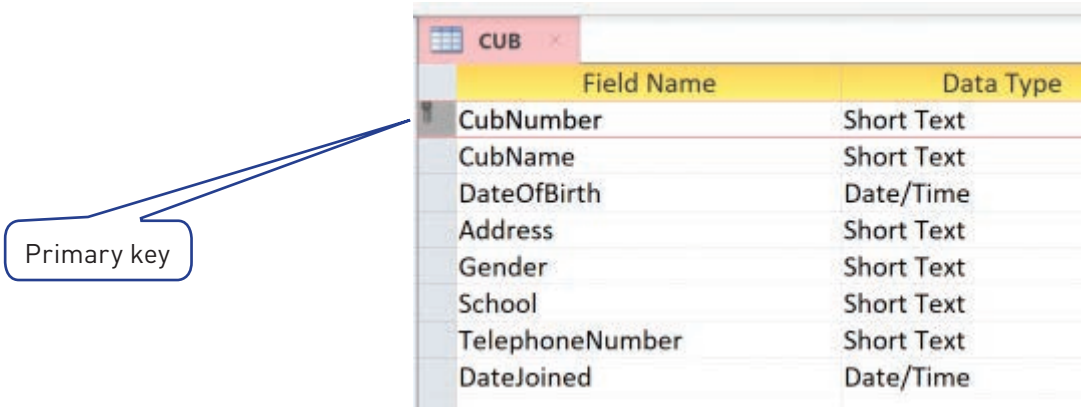
....and name the table **CUB**.



▲ **Figure 9.11** Naming the table

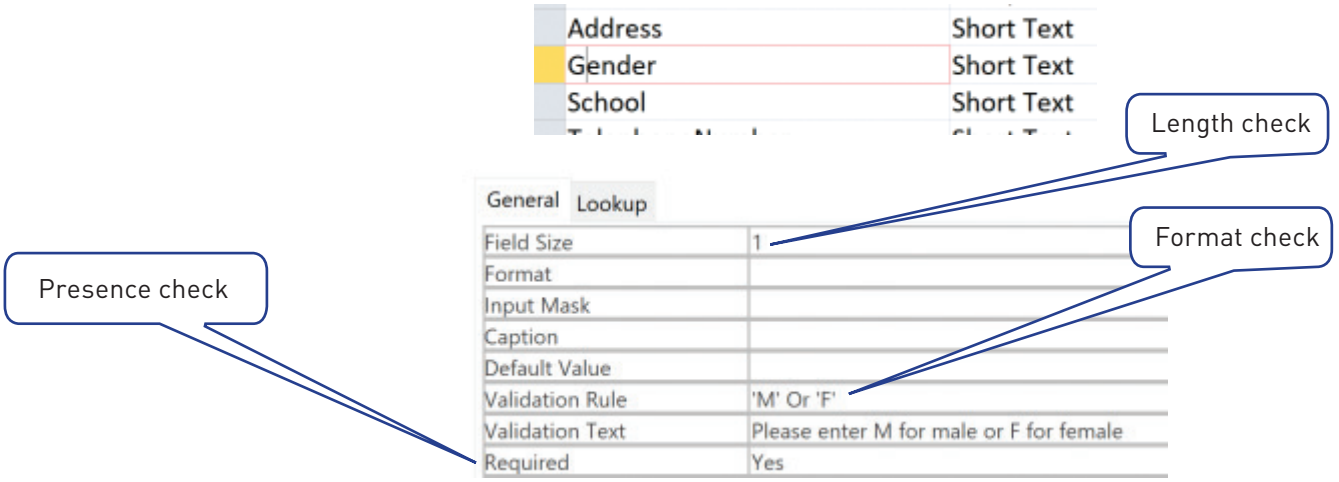
Set up the fields to match the data collection form in Figure 9.7 and include an extra field for a primary key.

Each field will require a meaningful name and a suitable data type must be selected.



▲ **Figure 9.12** Fields for the CUB table

Validation checks need to be built in for each field, for example the **Gender** field.



▲ **Figure 9.13** Validation rules for **Gender** field

Activity 9.7

1 **a** Set up a cub scout database including appropriate validation checks for each field.

Enter data for at least 10 records.

b The cub scout leader wants to put each cub into a group called a 'six'; each 'six' can have up to six cubs in it and is given a name for example red, yellow, blue and green.

Add a new text field called Six, and assign each cub to either a red, blue, yellow or green six.

i Write an SQL query to pick out any cubs in the red six.

ii Write an SQL query to pick out any cubs in the red six or the blue six.

iii Write an SQL query to count the number of cubs in the red six.

c The cub scout leader wants to calculate the number of badges that all the cubs have been awarded. Add a new integer field called *Badges*, enter the number of badges awarded to each cub.

Write an SQL query to count the number of badges awarded to the whole cub group.

Activity 9.8

1 A database of students is to be set up with the following fields:

- Family name
- Other names
- Student ID
- Date of Birth
- Date of Entry to School
- Current Class
- Current school year/grade
- Email address.

a Select a data type for each field.

b Which fields should be validated, and which fields should be verified?

c Decide the validation rules for those fields which should be validated.

d Which field would you choose for the primary key?

e Choose a suitable format for the student ID.

f Build a database with at least 10 records; include all your validation checks. Ensure there are at least 3 different classes and 2 different years/grades.

g Set up and test SQL scripts to:

i Display Other names, Family and Email address in alphabetical order of family name.

ii Select all the students from each class in alphabetical order.

iii Select all the students for each year/grade and print Other names, Family name and Date of Birth, grouping the students by class.

Link

For more on verification, see Chapter 7.